MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

12

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

Which expression represents the result of increasing a positive quantity w by 43%?

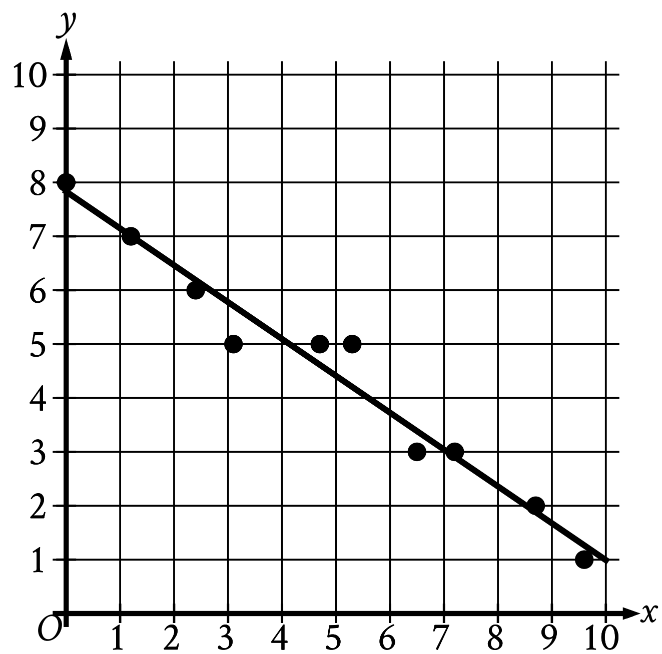
A. $1.43w$

B. $0.57w$

C. $43w$

D. $0.43w$

In the given scatterplot, a line of best fit for the data is shown.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- A line of best fit is shown:
- The line of best fit slants down from left to right.
    - 5 points are touching the line of best fit.
    - 2 points are above the line of best fit.
    - 3 points are below the line of best fit.
    - The line of best fit passes through the following approximate coordinates:
        - (1 comma 7)
        - (4 comma 5)
        - (10 comma 1)

Which of the following is closest to the slope of this line of best fit?

A. 7

B. 0.7

C. -0.7

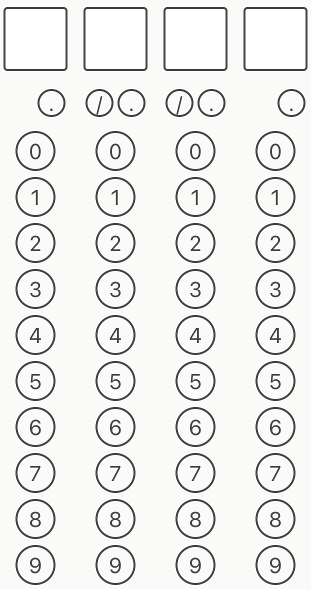
D. -7

Q3

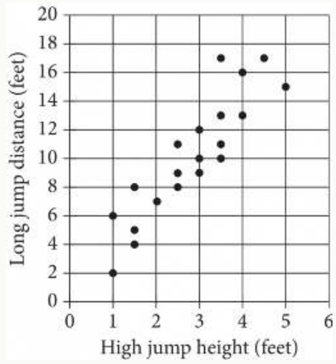
GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

On April 18, 1775, Paul Revere set off on his midnight ride from Charlestown to Lexington. If he had ridden straight to Lexington without stopping, he would have traveled 11 miles in 26 minutes. In such a ride, what would the average speed of his horse have been, to the nearest tenth of a mile per <u>hour</u>?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Each dot in the scatterplot above represents the height x, in feet, in the high jump, and the distance y, in feet, in the long jump, made by each student in a group of twenty students. The graph of which of the following equations is a line that most closely fits the data?

A. $y = 0.82x + 3.30$

B. $y = 0.82x - 0.82$

C. $y = 3.30x + 0.82$

D. $y = 3.30x - 3.30$

Q5

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

A competition consisted of four different events. One participant completed the first event with an average speed of 20.300 miles per hour. What was this average speed, in <u>yards</u> per hour? 1 mile $= 1{,}760$ yards

STUDENT-PRODUCED RESPONSE

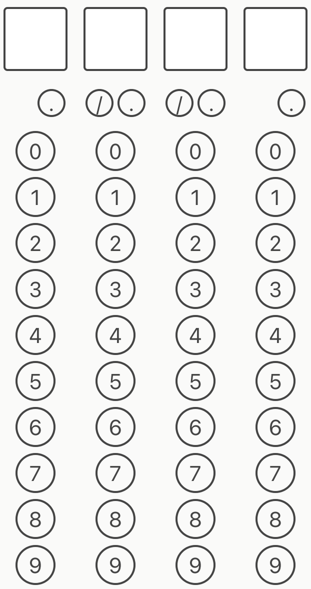
.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

One of a planet's moons orbits the planet every 252 days. A second moon orbits the planet every 287 days. How many more days does it take the second moon to orbit the planet 29 times than it takes the first moon to orbit the planet 29 times?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

Rectangle A has length 15 and width w. Rectangle B has length 20 and the same length-to-width ratio as rectangle A. What is the width of rectangle B in terms of w ?

A. $\frac{4}{3}w$

B. $w+5$

C. $\frac{3}{4}w$

D. $w-5$

Q9 PROBLEM-SOLVING AND DATA ANALYSIS

Which of the following speeds is equivalent to 90 kilometers per hour? (1 kilometer = 1,000 meters)

A. 25 meters per second

B. 32 meters per second

C. 250 meters per second

D. 324 meters per second

Q10 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

Pure beeswax has a density of 0.555 ounce per cubic inch. An online company sells pure beeswax at a price of $8.00 per ounce. What is the selling price, in dollars per cubic inch, for pure beeswax purchased from this company?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.